MIL-S-20386D(ME)
23 March 1977
SUPERSEDING
MIL-S-20386C(ME)
15 February 1967

MILITARY SPECIFICATION

STEREOMETER, PHOTOGRAMMETRIC:

STEREOCOMPARAGRAPH-TYPE;

WITH ACCESSORIES AND CASE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-illuminating sketching instrument consisting of a stereoscope, measuring system, drawing attachment, and alinement mechanism.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

DRAWINGS

ME

TA13214E6300	- Stereometer, Photogrammetric: Stereocomparagraph Type.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The stereometer shall be in accordance with Top Assembly TA13214E6300, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the stereometer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first-produced stereometer). The contractor shall furnish one or more stereometers as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce stereometers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

3.3 Material. Material shall be as shown on the applicable drawings.

3.4 Storage temperature. The stereometer shall not be damaged by storage in an ambient temperature of minus 65° F with no solar radiation for 36 hours, nor in an ambient temperature of plus 155° F for 4 hours.

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3.5 Items to be furnished with each stereometer. The following items in the quantities specified shall be furnished with each stereometer:

(a) Lamps - 2 each.
(b) Lead holders - 2 each.
(c) Lens, and retainer ring (floating mark lenses) - 2 each.
(d) Arm extension, 6-inch; with screw - 1 each.
(e) Types HB, 3H, and 5H black graphite leads - 1 dozen each.

3.6 Case. A fitted case shall be provided to house the stereometer and the items specified in 3.5. Padded blocks or cushions shall be provided to protect the instrument and its components from damage when tested as specified in 4.5.2.2. The case shall have a hinged cover with a clasp to hold the cover closed when the case is dropped from a height of 30 inches. The case shall be provided with a handle or strap for carrying. The following instructions in bold-type letters on plastic material shall be fastened on the inside of the cover of each case:

NOTICE

THE USE OF SOFT COTTON AND LIQUID CLEANER, SUCH AS AMMONIA WATER, IS RECOMMENDED FOR CLEANING THE FIRST SURFACE MIRRORS ON THIS INSTRUMENT. USE EXTREME CAUTION TO PREVENT DAMAGE TO THE MIRROR.

3.7 Identification marking. The stereometer shall be identified in accordance with MIL-STD-130. The marking shall be applied to the stereometer on plates conforming to MIL-P-514, Type III, material optional.

3.8 Workmanship. All parts, components, and assemblies of the stereometer including castings, molded parts, stamping, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, and other harmful extraneous material. All edges shall be rounded or chamfered. The reflecting surfaces of the mirror shall be clean and free from cracks, scratches, or pinholes. The mirror shall show no signs of wrinkling or peeling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first-produced stereometer shall be cause for disassembly, in the presence of a Government representative, of the first-produced stereometer to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first-produced stereometer. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.7 and 6.8.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced stereometer inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced stereometer inspection.

4.3.1 Examination. The stereometer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The stereometer shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Each stereometer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.3. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The stereometer shall be examined as specified herein for the following major defects:

101. Any part (or component) not in accordance with the SQAP requirements as shown on the drawings.
102. Any dimension, other than those identified on the SQAP requirements shown on the drawings, not as specified.
103. Components missing or not as specified.
104. Items to be furnished with each stereometer missing or not specified.
105. Case not as specified.
106. Identification marking missing or not as specified.
107. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Storage temperature. The case, with the stereometer and items specified in 3.5 in place, shall be placed in a temperature test chamber at normal room temperature. The temperature in the chamber shall be raised to plus 155° F and maintained at this temperature for 4 hours. The temperature shall then be allowed to return to normal room temperature, and the stereometer, items specified in 3.5, and case shall be examined for damage. The temperature in the chamber shall then be lowered to minus 65° F and maintained for 3 days. The chamber shall then be allowed to return to normal room temperature and the stereometer, items specified in 3.5, and case shall be examined for damage. Evidence of any damage shall constitute failure of this test.

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4.5.2.2 Drop. The case, with the stereometer and items specified in 3.5 in place, shall be dropped once on each corner (8 drops) from a height of 30 inches onto a solid wood floor. Any evidence of damage to the case or to the contents, or opening of the clasp, shall constitute failure of this test.

4.5.2.3 Mirror. The mirror tests shall be conducted before mirrors are installed in the instruments. Mirrors used in the tests for corrosion and abrasion resistance shall not be used in any instrument offered for acceptance by the Government. The tests for reflector factor shall be conducted prior to the tests for corrosion and abrasion resistance.

4.5.2.3.1 Reflection. The reflectivity test shall be in accordance with any standard optical method. The mirror shall be tested for reflection of light in the visible spectrum. Light reflection of less than 85 percent incident light shall constitute failure of this test.

4.5.2.3.2 Corrosion resistance. The mirror shall be completely immersed in a 6 percent salt (NaCl) solution at room temperature for a period of 8 hours, after which it shall be washed clean and closely examined for pinholes. It shall then be immersed again in the salt solution for a period of 8 hours and washed clean. Evidence of deterioration shall constitute failure of this test.

4.5.2.3.3 Abrasion resistance. A 3/4-inch-diameter steel ball covered with linen having a maximum of 60 threads per inch and weighted with a 500-gram weight shall be rubbed on the reflecting surface with a reciprocating motion for 100 cycles. Reduction of the reflectivity of the mirror's surface shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a stereometer prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

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108. Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
109. Box closure not in accordance with Method V as specified.
110. Blocking, bracing, and cushioning not provided as specified.
111. Strapping not as specified.
112. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each stereometer with repair parts, tools, and publications contained in the case shall be preserved in accordance with MIL-P-116, Method IIb. The case shall be the inner container, and the outer container shall be a box conforming to PPP-B-636, W5c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. Each stereometer shall be preserved to afford protection against deterioration and damage during shipment from the supply source to the initial destination.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Stereometers, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Blocking, bracing, and cushioning shall be provided to prevent movement within the box. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified for Level A except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. The stereometers, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking for commercial packaging shall reflect as a minimum, the item name, federal stock number, quantity, contract or purchase order number, and appropriate address markings. The marking shall be applied with a waterproof material, shall be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The stereometer is used to plot topographic data directly from stereoscopic pairs of aerial photographs.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first-produced stereometer and number of stereometers required (see 3.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When other than Finish B strapping is required (see 5.2.1).

6.3 Definitions and nomenclature. Definitions and nomenclature of terms used in this specification may be found in MIL-STD-150, MIL-STD-1241, MIL-HDBK-25, and MIL-HDBK-141.

6.4 First-produced stereometer. Any changes or deviations of production stereometers from the approved first-produced stereometers during production will be subject to the approval of the contracting officer. Approval of the first-produced stereometer will not relieve the contractor of his obligation to furnish stereometers conforming to this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each stereometer.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of stereometers (see 3.5 for items included with each stereometer).

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6.7 Supplementary quality assurance provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.8 Definition.

6.8.1 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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DOCUMENT IDENTIFIER AND TITLE MIL-S-20386D(ME) Stereometer, Photogrammetric: Stereocomparagraph-Type; with Accessories and Case

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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